



Knee extensor torque-velocity relationships following anterior cruciate ligament reconstruction

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ABSTRACT

Background: The inherent nature of the torque-velocity relationship is the inverse nature between the velocity of muscle contraction and torque production and is an indication of muscle function. The purpose of this study was to characterize the torque-velocity relationship in the quadriceps following anterior cruciate ligament reconstruction compared to healthy limbs.

Methods: 681 participants were included, 493 of which were patients at least four months following anterior cruciate ligament reconstruction (23.2 ± 10.08 yr, 6.6 ± 5.37 months post-surgery) and 188 were healthy participants (21.6 ± 3.77 yr). A subset of 175 post-surgical participants completed a repeated visit (8.1 ± 1.71 months post-surgery). Participants completed isokinetic knee extension at $90^\circ/\text{s}$ and $180^\circ/\text{s}$. A one-way ANOVA was used to compare torque velocity relationships by limb type (surgical, contralateral, healthy). Paired samples *t*-tests were conducted to analyze the torque-velocity relationship across limbs and across time.

Findings: There was a large effect for limb type on torque-velocity ($F_{(2, 1173)} = 146.08$, $p < 0.001$, $\eta^2 = 0.20$). Surgical limbs demonstrated significantly lower torque-velocity relationships compared to the contralateral limbs (ACLR: 0.26 Nm/kg, contralateral: 0.55 Nm/kg, $p < 0.001$, $d = 1.18$). Healthy limbs had similar torque-velocity relationships bilaterally (dominant limb: 0.48 Nm/kg, non-dominant limb: 0.49 Nm/kg, $p = 0.45$). The torque velocity relationship for the involved limb significantly increased in magnitude over time ($+0.11$ Nm/kg, $p < 0.001$, $d = -0.61$) while the contralateral limb torque-velocity relationship remained stable over time (0.0 Nm/kg difference, $p = 0.60$).

Interpretation: Following surgery, the knee extensors appear to have altered torque-velocity relationships compared to contralateral and healthy limbs. This may indicate a specific target for assessment and rehabilitation following surgery.

1. Introduction

Injuries to the anterior cruciate ligament (ACL) often lead to time loss from physical activity and decreased patient quality of life (Stewart et al., 2017). Surgical intervention is the current gold standard of treatment and 100,000 s of ACL reconstructions (ACLR) occur in the US annually (Krause et al., 2018; Stewart et al., 2017). Post-surgical

treatment consists of a lengthy rehabilitation process in order to restore preoperative function, however among patients who return to previous levels of activity following ACLR, reinjury rates remain high (Barber-Westin and Noyes, 2020). Before allowing patients to return to high level pre-injury activities such as cutting and pivoting sports, objective strength testing is typically included in the post-injury decision making process in order to reduce risk of reinjury (Webster and Feller,

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2020; Webster and Hewett, 2019). While strength test performance has been related to joint loading and patient self-reported function, muscle strength is often characterized by calculating limb symmetry of the involved (i.e. surgically reconstructed) limb compared to the uninvolved (contralateral) limb at isolated contraction velocities (Garrison et al., 2019; Norte et al., 2018). While testing at a single velocity may suffice in establishing muscle capacity muscle function is derived from numerous characteristics including the torque-velocity relationship (Alcazar et al., 2019).

The clinical utility of side-to-side symmetry has been critiqued because the contralateral limb undergoes post-surgical changes in strength and may not be a stable comparator (Wellsandt et al., 2017). Traditional measures of limb symmetry and strength attempt to characterize deficits following injury, but magnitude of strength alone may not elucidate other changes in muscle function. Injury to the ACL has implications for quadriceps activation, as neural pathways may inhibit muscle strength post-injury (Hart et al., 2010; Lepley et al., 2015). While strength of the surgical limb may be reduced due to factors including use of nerve blocks, tourniquet use during surgery, and altered movement patterns following surgery, gross strength and symmetry only provide limited information about the complex properties of skeletal muscle (Baron et al., 2020; Gardinier et al., 2012). At current there is no valid unilateral measure that provides an alternative to our current use of limb symmetry.

The force-velocity relationship of skeletal muscle is an inverse relationship between the velocity of muscle contraction and force production (Alcazar et al., 2019; Bohm et al., 2018). This relationship has been established in single muscle fibers, animal models, and in healthy human muscles (Edman, 1988; Edman, 2005; Gillen et al., 2020; Hill, 1938; Pain et al., 2013). The torque-velocity relationship follows the same principles as the force velocity relationship and is used to characterize muscle function in human subjects (Gillen et al., 2020; Pain et al., 2013). Descriptions of either force-velocity or torque-velocity relationships in literature are often determined by exercise selection or by conversion of torque into linear force (Janicijevic et al., 2020; Pain et al., 2013). In healthy individuals examining knee extensor muscles, the torque-velocity relationship has been characterized by higher peak torques at lower contraction velocities indicating uninjured skeletal muscle (Gillen et al., 2020; Janicijevic et al., 2020; Pain et al., 2013). Greater differences in torque production across velocities has been associated with higher activity and training levels and is seen as a positive adaptation of the torque-velocity relationship (Janicijevic et al., 2020; Sasek et al., 2022). It is unknown whether knee injury-related muscle weakness may alter this inherent physiologic property of skeletal muscle, and thereby may have implications in evaluating rehabilitation progress following ACLR. Therefore, the primary purpose of this study was to determine the nature of the torque-velocity relationship following ACLR. A secondary purpose was to determine if the torque-velocity relationship changed over time with serial assessments in patients recovering from ACLR. We hypothesized that following ACLR, participants would display smaller torque-velocity relationships, indicated by smaller differences in peak torque production across velocities, compared to both the contralateral limb and healthy controls. We also anticipate that this difference would decrease over time as participants continued post-surgical rehabilitation.

2. Materials

This was a prospective cohort study including a longitudinal follow-up in a subset of patients post-ACLR. This study included participants following ACLR (242 male, 252 female) and healthy participants (75 male, 113 female). Participants with ACLR were included if they had a history of a primary, unilateral, and uncomplicated ACLR at least four months prior to testing and excluded if they had a lower-extremity joint surgery prior to the index ACLR, multiple ligament reconstruction, history of graft failure, contralateral knee surgery, lower-extremity

injury within 6 months, or we could not conform prior surgical history from medical records review. Participants were all given standardized rehabilitation protocols by their surgeon, but adherence was not monitored. All participants completed testing during a single session (Visit 1); a subset of ACLR participants returned for optional repeated follow-up testing (Visit 2) approximately two months following Visit 1. All participants were given the option to attend a second visit, but those who elected not to return were still included in the analysis. Independent variables included contraction velocity (90°/s, 180°/s), and time (Visit 1, Visit 2) for those who completed more than one visit. The dependent variable of interest was a torque velocity variable created by the difference in peak torque production in newton meters (Nm). This difference (Peak Torque generated at 90°/s - Peak Torque generated at 180°/s) is representative of the same value as a slope that would be derived from a linear equation (Janicijevic et al., 2019). This study was approved by the institutional review board and all participants provided informed consent prior to enrollment. ACLR participant inclusion can be seen in Fig. 1.

2.1. Strength testing

Before beginning strength testing participants completed a five minute treadmill walk. Knee extensor strength was evaluated using a multi-mode dynamometer (Biodex Inc. Shirley, NY) as previously described (Norte et al., 2019). Participants were seated in approximately 85° of hip flexion and 90° of knee flexion with their head and back against the back of the dynamometer chair. All participants were instructed to keep their arms crossed across their chest during the trials. Each participant exerted maximal effort through their full range of motion for eight consecutive trials of isokinetic concentric knee extension at 90°/s and 180°/s, beginning with 90°/s. Before completing the test trials participants were allowed to complete practice trials to familiarize themselves with the testing technique and velocity in order to make sure that they were comfortable with the task. Participants were given visual and verbal feedback throughout each trial.

2.2. Data Processing

The knee extension peak torque (Nm) was recorded and normalized by participants' body mass (kg). The torque-velocity relationship was operationally defined as the difference in normalized peak torque measured at 90°/s and 180°/s. Measures were collected bilaterally, starting with the contralateral limb for ACLR participants and the non-dominant limb for healthy participants.

2.3. Statistical analysis

For the primary aim (Visit 1) a one-way ANCOVA by graft type was run to compare torque-velocity for ACLR participants when including time post-surgery as a covariate. An additional one-way ANOVA was used to compare peak torque-velocity across limb types (ACLR involved, ACLR uninvolved, Healthy control). Post hoc paired *t*-tests were performed to determine interlimb differences at Visit 1. For the secondary aim (Visit 2) paired samples *t*-tests were conducted to analyze the torque-velocity relationship across limbs and across time. A preliminary analysis revealed that, when looking for a minimal difference of 0.3 and a standard deviation for 0.47 based on pilot data collected from this sample, performing the statistical analysis with a power of 0.80, $\alpha = 0.05$, and a 20% subject attrition rate should include at least 47 participants. Statistical analyses were performed in SPSS Version 27.0 and Stata version 17.0 (IBM Corp, 2020; StataCorp, 2021). Alpha level was set a priori at $\alpha < 0.05$. Effect sizes for multivariate testing were interpreted as small ($\eta^2 = 0.01$), medium ($\eta^2 = 0.06$), and large ($\eta^2 = 0.14$) effects. Effect sizes for post hoc analyses were reported via Cohen's *d* and interpreted as follows: $d = 0.2$ small, $d = 0.5$ medium, and $d = 0.8$ large (Lakens, 2013).

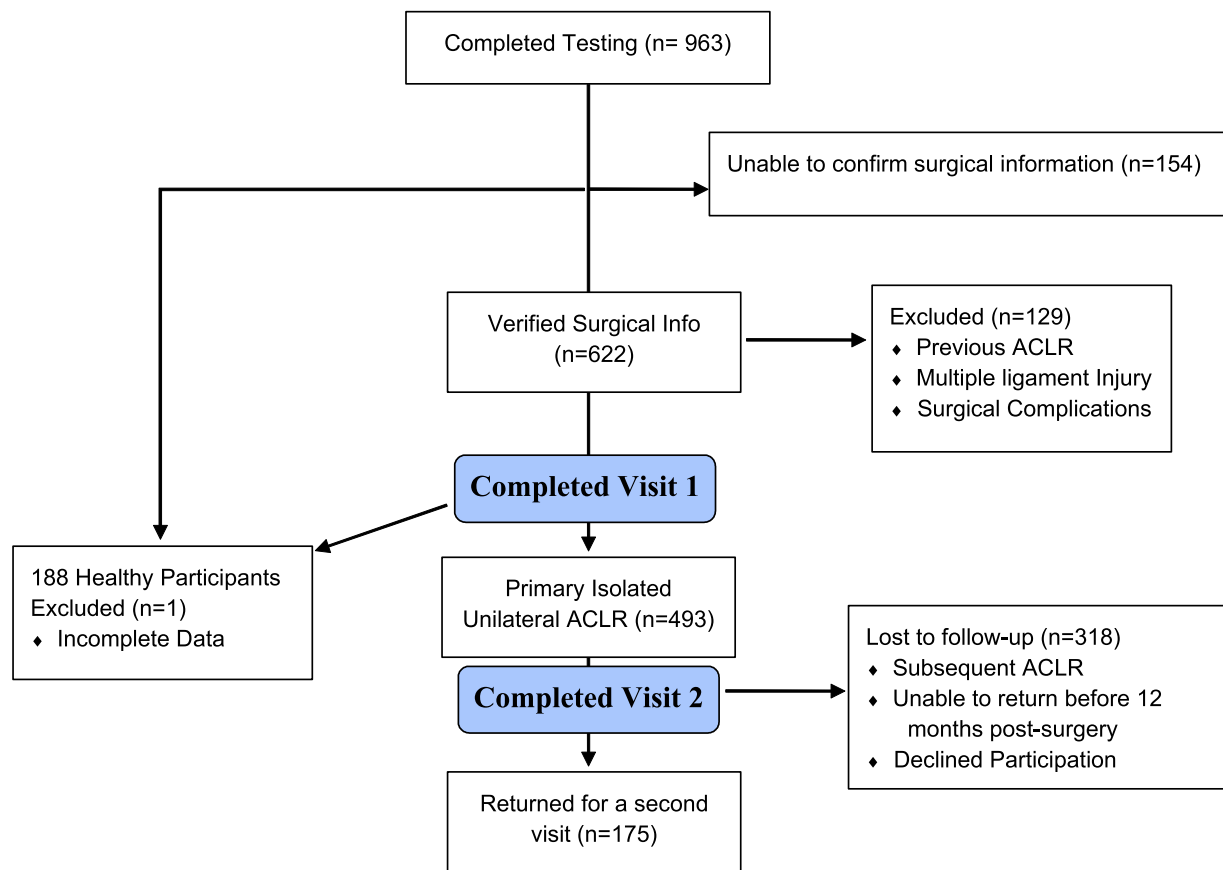


Fig. 1. Consort Diagram showing Inclusion and Exclusion for each group and visit included in this study.

3. Results

282 participants were excluded from analysis based on our pre-determined exclusion criteria, leaving us with 681 participants used for analysis. 681 participants completed Visit 1 (ACLR: $n = 493$; Healthy Control: $n = 188$). Descriptive information for these patients can be found in Table 1. A subset of 175 ACLR participants completed both Visit 1 and Visit 2.

Table 1
Participant Demographics.

	ACLR ($n = 493$)	Healthy ($n = 188$)
Aim 1 Participants ($n = 681$)		
Age (yr)	23.2 ± 10.08	21.6 ± 3.77
Sex	242 M, 252F	75M, 113F
Height (cm)	172.5 ± 12.16	172.6 ± 10.28
Mass (kg)	75.9 ± 18.02	69.9 ± 12.28
Time post-surgery	6.6 ± 5.37 months	
Graft Type N (%)	320 patellar tendon (64.9%) 146 hamstring (29.6%) 2 hybrid (0.4%) 6 quadriceps tendon (1.2%) 1 allograft (0.2%) 18 unknown (3.7%)	
Aim 2 Participants ($n = 175$)		
Age (yr)	22.5 ± 9.42	
Sex	78 M, 97F	
Height (cm)	172.5 ± 10.14	
Mass (kg)	77.2 ± 18.02	
Time post-surgery	5.23 ± 1.25 months (Visit 1), 8.1 ± 1.71 months (Visit 2)	
Graft Type N (%)	131 patellar tendon (74.9%) 42 hamstring (24.0%) 2 quadriceps tendon (1.1%)	

For ACLR participants with a single visit there was no significant association of graft type on the torque-velocity relationship ($p = 0.07$). There was a large effect for limb type on torque-velocity ($F_{(2, 1173)} = 146.08$, $p < 0.001$, $\eta^2 = 0.20$) at 6.6 ± 5.37 months post-surgery. The differences in peak torque production can be seen in Fig. 2. Post hoc analysis revealed that the ACLR limbs demonstrated significantly lower torque-velocity relationships compared to the contralateral limbs (ACLR: 0.26 Nm/kg, contralateral: 0.55 Nm/kg, $p < 0.001$, $d = 1.18$). Healthy limbs had similar torque-velocity relationships bilaterally (dominant limb: 0.48 Nm/kg, non-dominant limb: 0.49 Nm/kg, $p = 0.45$).

For participants who completed two post-operative visits, the torque-velocity relationship of the involved limb was significantly different from the contralateral limb at Visit 1 (ACLR: 0.21 Nm/kg, contralateral: 0.56 Nm/kg, $p < 0.001$, $d = -1.40$) and at Visit 2 (ACLR: 0.32 Nm/kg, contralateral: 0.56 Nm/kg, $p < 0.001$, $d = -1.09$). The torque velocity relationship for the involved limb significantly increased in magnitude over time ($+0.11$ Nm/kg, $p < 0.001$, $d = -0.61$, Fig. 3). Contralateral limb torque-velocity relationship remained stable over time (0.0 Nm/kg difference, $p = 0.60$). Peak torque production at each velocity across visits can be found in Table 2.

4. Discussion

The primary findings of this study found following ACLR, participants exhibited altered torque-velocity relationships within the knee extensors compared to contralateral limbs and healthy controls. The difference between the knee extensor torque-velocity relationship in ACLR and contralateral limbs decreased over time, but the torque-velocity relationship of the ACLR limbs remained statistically different from the relationships observed in contralateral limbs and healthy controls, as demonstrated by smaller differences in peak torque

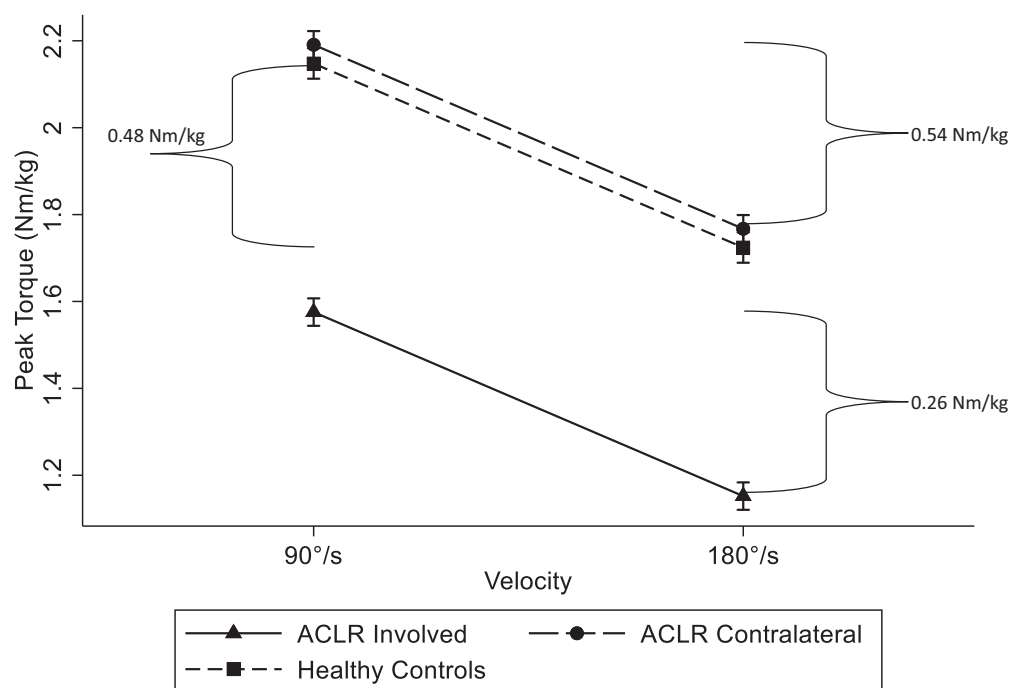


Fig. 2. Torque- Velocity Relationships across Limb Types w/ 95% confidence intervals.

*Torque-velocity relationship significantly different from Anterior Cruciate Ligament (ACLR) Involved limbs.

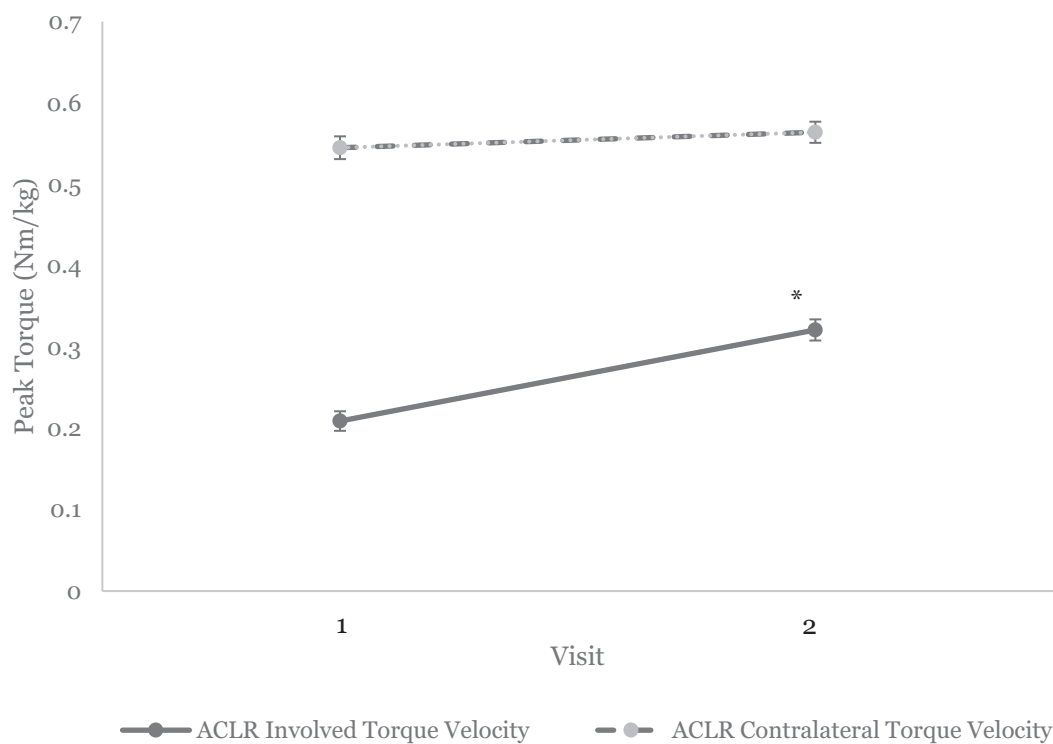


Fig. 3. Torque- Velocity Relationships by Limb Type across Time w/ 95% confidence intervals.

*Torque-velocity relationship significantly different from Visit 1.

Anterior Cruciate Ligament (ACLR) Involved Visit 1 Peak Torque Difference = 0.21 Nm/kg, Visit 2 Peak Torque Difference = 0.32 Nm/kg.

Anterior Cruciate Ligament (ACLR) Contralateral Visit 1 Peak Torque Difference = 0.54 Nm/kg, Visit 2 Peak Torque Difference = 0.56 Nm/kg.

production across velocities. These findings support our hypothesis that the torque-velocity relationship is altered after ACLR and can change over time in the recovery period. It is expected that as contraction velocity increased peak torque would decrease (Pain et al., 2013). This pattern was observed in all limb types as evidenced by the main effect

for limb type, but the mean difference in torque production across velocities was smaller in magnitude for ACLR involved limbs at approximately 6.6 months post-surgery as evidenced by the main effect for velocity. The ACLR limbs were weakest at all angular velocities, compared to contralateral and healthy limbs, and had the smallest peak

Table 2

Normalized Peak Torque (Nm/kg) at each Contraction Velocity (n = 681).

Limb Type	ACLR (Involved Limb) n = 493	ACLR (Contralateral Limb) n = 493	Healthy (Average of Both Limbs) n = 188
Isokinetic PT @ 90°/s	1.5 ± 0.50	2.3 ± 0.47*	2.2 ± 0.44*
Isokinetic PT @ 180°/s	1.2 ± 0.38	1.7 ± 0.38*	1.7 ± 0.37*

Normalized Peak Torque (Nm/kg) at each Contraction Velocity across Time (n = 175)

Limb Type	ACLR (Involved Limb) n = 175		ACLR (Contralateral Limb) n = 175	
	Visit 1	Visit 2	Visit 1	Visit 2
Isokinetic PT @ 90°/s	1.4 ± 0.44	1.7 ± 0.46**	2.3 ± 0.46	2.4 ± 0.45
Isokinetic PT @ 180°/s	1.2 ± 0.35	1.4 ± 0.37**	1.8 ± 0.38	1.8 ± 0.36

* Significantly different from ACLR involved limb ($p < 0.001$).** Significantly different from Visit 1 ($p < 0.001$).

torque difference between 90°/s and 180°/s indicating low variability in peak torque production (see Fig. 2). This lack of variability may be attributed to the lower maximal peak torque outputs in the ACLR limb at the slowest testing velocity compared to contralateral and healthy limbs (Table 2). This is supported by the increase in variability over time that was accompanied by an increase in peak torque at 90°/s in participants that completed multiple visits. Participants that completed repeated visits produced greater peak torque at both contraction velocities, and a greater difference between peak torque across velocities at their second visit in ACLR involved limbs (see Fig. 3).

Previous research has characterized the force velocity and torque-velocity relationships as linear, hyperbolic, or double-hyperbolic curves, however when evaluating during single-joint movements, the torque relationship has been characterized as a linear relationship using a two point method (Alcazar et al., 2019; Janicijevic et al., 2019; Janicijevic et al., 2020). Some prior studies characterized the torque-velocity relationship using linear regression to calculate a slope representative of the difference in peak torque output across velocities (Janicijevic et al., 2019; Janicijevic et al., 2020). In the current study we aimed to compare peak torque between velocities as a first step in characterizing the torque-velocity relationship in patients recovering from ACLR using metrics that are traditionally used and recognizable. The focus of future work will leverage more advance analytics to further develop theory in this area by characterizing this relationship with further datapoints.

Peak torque, as measured by a dynamometer during isokinetic or isometric contractions, is often used to quantify muscle strength and symmetry to make comparisons to the contralateral limb (Webster and Hewett, 2019). The magnitude of change in the peak torque production across contraction velocities is an indicator of a change in torque-velocity relationships and may be characteristic of changes in muscular properties (Labarque et al., 2002). Previous research including healthy participants, found that knee extensor torque-velocity relationships were altered after two weeks of immobilization which was hypothesized to be representative of changes in muscle architecture (Labarque et al., 2002). A systematic review and meta-analysis of morphological characteristics of the vastus lateralis demonstrated that the pooled type II muscle fibers sustain more atrophy in terms of cross-sectional area than type I fibers over an average period of disuse of four weeks (Vikne et al., 2020). Given the limited ability of patients to utilize the quadriceps following surgery, it is likely that similar sequelae are occurring in ACLR patients.

Similar to our findings, others have reported force and torque-velocity relationship differences to be primarily driven by the maximum capacity at slower velocities (Andersen et al., 2005; Cuk et al.,

2016; Janicijevic et al., 2020). At slower contraction velocities, peak torque production may be more directly influenced by fiber type distribution and size with greater peak torque expected to be associated with greater relative size of fast twitch fibers (Gregor et al., 1979; Tihanyi et al., 1982). Previously reported muscle fiber type composition changes following ACLR include decreases in fast twitch type-IIA fiber frequency and increases in type-IIA/X hybrid fibers in the involved limb that may be corrected with targeted strength training (Friedmann-Bette et al., 2018; Noehren et al., 2016). These fibers may be targeted with interventions including high loads, high velocity contractions, and eccentric exercise (Friedmann-Bette et al., 2018; Tegtbur et al., 2009; Wilson et al., 2012). Muscle biopsies are considered the gold standard of direct measurement of skeletal muscle tissue properties however the procedure is invasive and has limited ability to capture dynamic muscular activities (Ward et al., 2009; Young et al., 2017). Previous research found correlations ranging from 0.57 to 0.77 between fiber type composition and isokinetic torque production in angular velocities ranging from 45 to 225°/s in male participants, but biopsies appear inappropriate to capture changes in torque-velocity relationships through serial assessment over short time periods (Froese and Houston, 1985; Labarque et al., 2002).

Further exploration specifically into potential torque-velocity changes following injury should be conducted in contrast to healthy participants with control for similar activity levels and training experience. Additional work should evaluate the torque-velocity relationship of muscle groups in unison with direct measures of muscle physiology.

Despite limitations in limb symmetry measures, the heterogeneity of reported torque-velocity relationships make the comparison of the involved and contralateral limbs a necessity for individuals post-ACLR due to patient specific differences (Wellsandt et al., 2017). In a sample of elite athletes, age, gender, and the primary event that participants trained for all affected the force velocity relationship, but the influence of previous sport experience and recent training exposures were not considered in the present study (Schleicherdt et al., 2019). In this sample, the torque-velocity in the surgical limbs appeared to approach the torque-velocity characteristics of the contralateral limbs over time, while the contralateral limbs remained stable. The torque-velocity relationship of muscle groups can be altered through intervention and as such should be a consideration for the restoration of limb symmetry (Labarque et al., 2002). While we did not directly examine training interventions in this sample, all participants were following a rehabilitation protocol that encouraged high velocity running, agility training, and plyometrics as appropriate beginning 4–6 months post-surgery, suggesting that by approximately 8 months post-surgery they should have all experienced these interventions. Longitudinal analysis of torque-velocity relationships may elucidate changes in muscular properties over time and allow comparison to the muscular properties of contralateral limb beyond maximal strength capability. The effect of traditional rehabilitation models and the role of training stimuli must also be explored ACLR.

Patients appear to have differences in the torque-velocity relationship in their ACLR limb on average 6.6 months following surgical intervention. In the current study, the torque-velocity relationship of ACLR limbs changed after, on average 2.9 months between repeat measurements, with high peak torque outputs at each velocity and a greater difference in peak torque across velocities. This relationship remained stable in ACLR patients' contralateral limbs as the change in mean difference in torque production was on average 0.02 Nm/kg at the latter visit, demonstrating the torque-velocity relationship may be altered in a single limb within the same individual. The torque-velocity relationship of the ACLR limbs at Visit 2 was closer to that of contralateral limbs and healthy controls than at Visit 1. Despite this change, the relationship remained altered compared to the non-injured limbs.

Based on the findings of the current study, clinicians who have access to serial isokinetic knee extensor strength testing may pay closer attention to how peak torque changes over time between multiple

contraction velocities. The decreased torque-velocity relationship relative to the contralateral limb may limit or alter the function of the knee extensors (Janicijevic et al., 2020). The clinical implications of these differences may become more pronounced during activities that place high external load on the knee or require high contraction velocities, such as activities of sport (Sašek et al., 2022). Potential interventions that target fast twitch muscles and may shift the torque-velocity relationship include exercise with high external loads, high velocity contractions, and/or eccentric exercise (Friedmann-Bette et al., 2018; Tegtbur et al., 2009; Wilson et al., 2012). It is currently unknown how addressing torque-velocity relationships will affect long term surgical outcomes or how this relationship continues to change beyond 8 months post-surgery. Future research should attempt to determine optimal torque-velocity relationships post-ACLR.

4.1. Limitations

This investigation was not without limitations. Patients referred to participate were all from a single academic center however from different surgeons, different rehabilitation protocols, previous levels of sport, and non-uniformly distributed graft types. As this was a point of care study the velocities were consistent with the clinical assessments patients were currently undergoing. We cannot speculate about how these results would change with a more homogenous sample, but we believe that this paper has strong external validity. Future analysis of torque-velocity relationships may need to be personalized with consideration to patient history. We are unsure how rehab compliance, sex, or previous exercise participation may affect this relationship. While it was assumed that there were no bilateral differences in torque-velocity relationships before surgical reconstruction, this could not be confirmed.

5. Conclusions

While the inherent nature of the torque-velocity relationship in the knee extensors of patients with primary, isolated ACLR was similar, the magnitude of was altered when compared to both contralateral and healthy control limbs. This relationship did change over time in the post operative period following ACLR to mirror contralateral and healthy limbs more closely but remained altered. Testing across multiple contraction velocities to evaluate the torque-velocity relationship may be able to provide insights that evaluating peak torque in isolation. This paper is the first step in establishing torque-velocity changes following ACLR and the future implications for reinjury should be explored.

Declaration of Competing Interest

The authors report there are no competing interests to declare.

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